

RP3. User involvement and participatory design

Status	Done
Date	2026/04/20 19:00
Topic	Design and innovation processes - Scripts and configuring users • Social construction of technology • Socio-technical imaginaries • Scripts and configuring users
Importance	⚠️ High Effort – High Value
Understanding	🙂 Good
URL	https://canvas.kth.se/courses/62723/assignments/370782
Teacher(s)	👩‍🏫 Lucy McCarren
Format	🪑 On-site
📒 Course Offering	👩‍💻 CM2010 VT26-4 Sociocultural Perspectives to Innovative Technologies 7.5 credits
Parent item	📚 CM2010 Lecture 3

The purpose of the short reaction paper is to summarise, analyse and reflect the content of the lecture in relation to own field of study, user study or interests. Short reaction paper consists of 1) introducing and summarising the main content of the lecture, and presenting and explaining key concepts 2) discussing the ideas and arguments presenting in the lecture and 3) connecting the content of the lecture to your own field of study, user study or interests. The length of the short reaction paper is 1-2 pages (500-1000 words).

On Tuesday, March 31, the third lecture in the course CM2010, held by Lucy McCarren, continued on the discussion of technology as something that is not value-neutral. Examples like low overpass bridges, nuclear power, seatbelt testing, and facial recognition showed how artifacts can embed politics, assumptions, exclusion, and power relations. The lecture also highlighted three main theoretical frameworks: socio-technical imaginaries, semiotic approaches to users, and social constructivism. First, socio-technical imaginaries are shared visions of desirable futures shaped by science, society, and technology. Second, semiotic approaches to users focus on what is built into the artifact itself and how technologies communicate with users through design. Third, social constructivism, or Social Construction of Technology (SCOT), argues technology is socially constructed, not determined solely by technical logic.

A central idea within Science & Technology Studies (STS) is that technology and society shape one another mutually, rather than technology developing independently and only later influencing society. From this perspective, the lecture made it clear that the notion of value-neutral technology is difficult to defend. Design always involves judgments about what is good and what is bad to determine whether something is desirable, useful, efficient, safe, normal, etc. It, therefore, makes assumptions about the people and contexts for which a technology is intended. The cases discussed in the lecture demonstrate this clearly, as technologies often benefit some groups while disadvantaging others. In the example of low overpass bridges, the infrastructure preventing buses to pass while favouring wealthier groups with private cars access to certain areas, thereby reinforcing social exclusion. Whether these effects are deliberate or unintended, they show how technological design can embody social assumptions and political

consequences. Thus, technology, through science, shapes society, and vice versa.

At the broader level, socio-technical imaginaries explain how society collectively envisions desirable futures. They do not direct individual design choices in a simple or mechanical way, but link technological possibilities with social goals. Although they reflect socio-cultural values, norms, and expectations, they do not remain abstract; instead, they shape innovation, research agendas, and policy priorities, as they influence how designers imagine future users and the environments in which technologies operate. For example, if the dominant vision holds that technology should make people more efficient, safe, or healthy, designers are likely to create systems aimed at target groups who fit that imagined future.

At the smaller level, semiotic approaches examine how assumptions and expectations become built into artifacts through user representations, inscriptions, and scripts. A user representation is the imagined version of the future user. An inscription is the built-in design feature. A script, or scenario, is the final framework of logic and actions that come out of that design, shaping behaviour by enabling some actions and discouraging others. From an STS perspective, artifacts “communicate” instructions, norms, and assumptions to users through these embedded components. As a result, the final product imagines a certain kind of users and, through its use, helps configure that user’s behaviour. This further demonstrates that technology is not value-neutral.

One part of the lecture I found particularly interesting was the discussion of social constructivism (SCOT), especially the way different social groups negotiate what a technology eventually becomes. In SCOT, technologies are shaped by relevant social groups, by interpretive flexibility, and by eventual closure and stabilization through negotiation. This means that technologies do not have fixed meanings from the outset; instead, their meaning and form emerge through the interactions of different users with different interests and perspectives.

In my opinion, while I was sitting in the lecture, the content was more difficult to grasp than in the previous lecture, partly because the two are so closely related. Even after reviewing the lecture again, it was not immediately clear to me that its main focus was the movement from the broader social level to the more specific

level of design. However, after spending more time with the material, I began to understand this connection, which made the lecture much more interesting.